

WASP-43b

R-0457

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_240207 · created 2026-07-19T19:17:20+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460347.8932 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.056
Transit depth (Rp/Rs)^2	3.84 +/- 2.21 [%]
Orbital Inclination (inc)	80.47 +/- 2.89
Airmass coefficient 1 (a1)	0.75 +/- 0.41
Airmass coefficient 2 (a2)	0.16 +/- 1.08
Scatter in the residuals of the lightcurve fit is	51.54 %
Best Comparison Star	None
Optimal Aperture	2.96
Optimal Annulus	3.36
Transit Duration (day)	0.044 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.056 vs 0.1594 (deviation 0.6σ)
Duration fit vs published	0.044 d vs 0.0512 d (deviation 0.44σ)
Mid-time O–C	6.0 min
Depth SNR	3.2
Residual scatter	51.54 %

Light curve

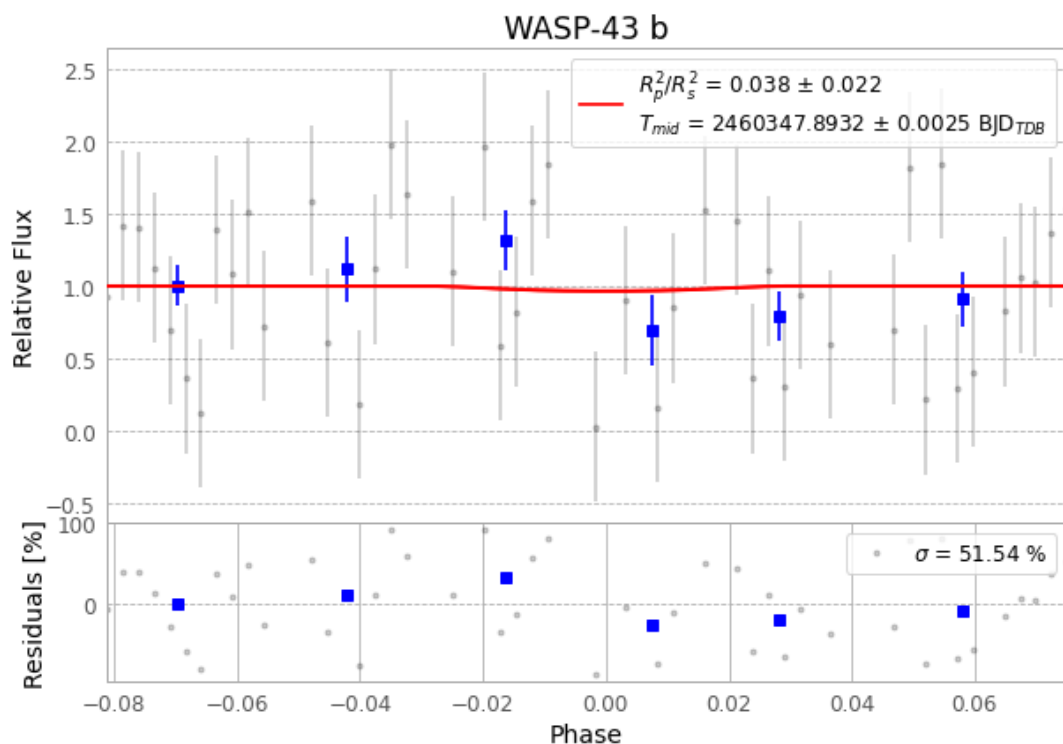


Figure 1 — FinalLightCurve_WASP-43 b_2024-02-07.png (SHA-256 6e906b701b00df38...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [428, 140], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 256b2c9eb737167f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

64 FITS frames (42 MB), WASP-43240207073915.FITS → WASP-43240207104815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-240207.log (90e03ce0c8ec...), FinalLightCurve_WASP-43 b_2024-02-07.png (6e906b701b00...), FinalLightCurve_WASP-43 b_2024-02-07.csv (3c7a97e79096...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 19:17Z.